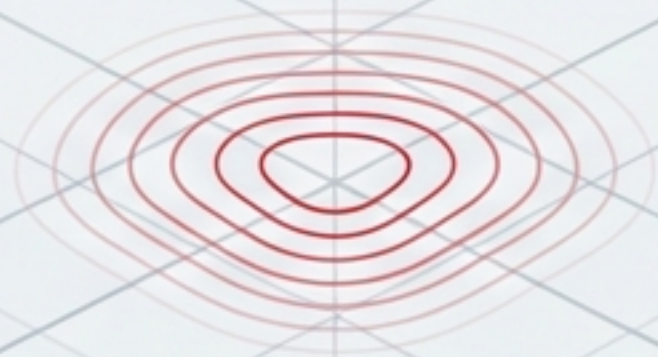


The Anatomy of a DrDoS Attack

How Attackers Turn the Internet Against Itself

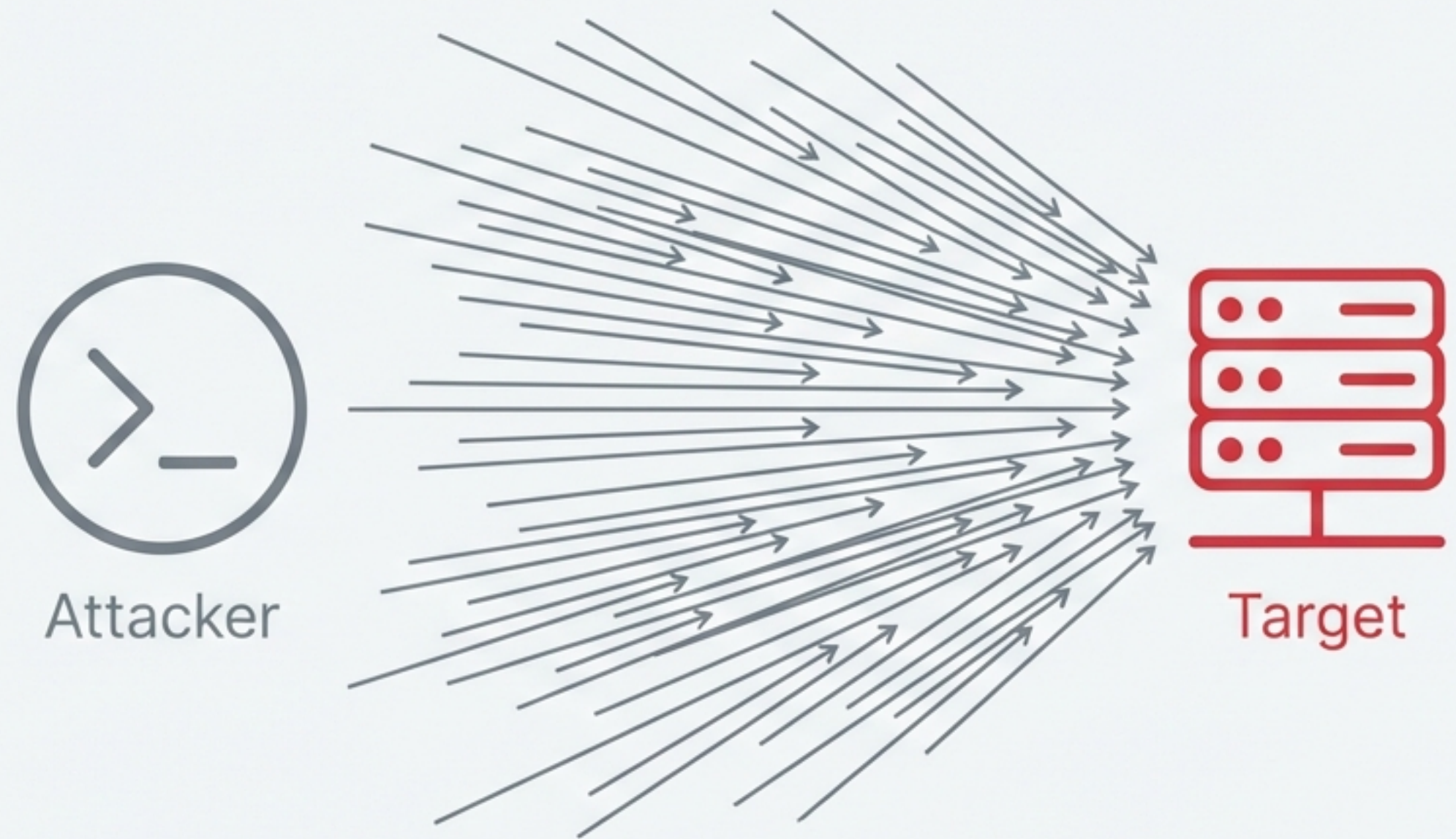


The Foundational Threat: The Denial-of-Service (DoS) Attack

A Denial-of-Service attack has one primary objective: to make a target system or network unavailable to its intended users.

The attacker achieves this by overwhelming the target with a flood of traffic or malicious requests.

This consumes the target's computational resources or network bandwidth, effectively shutting it down for legitimate traffic.

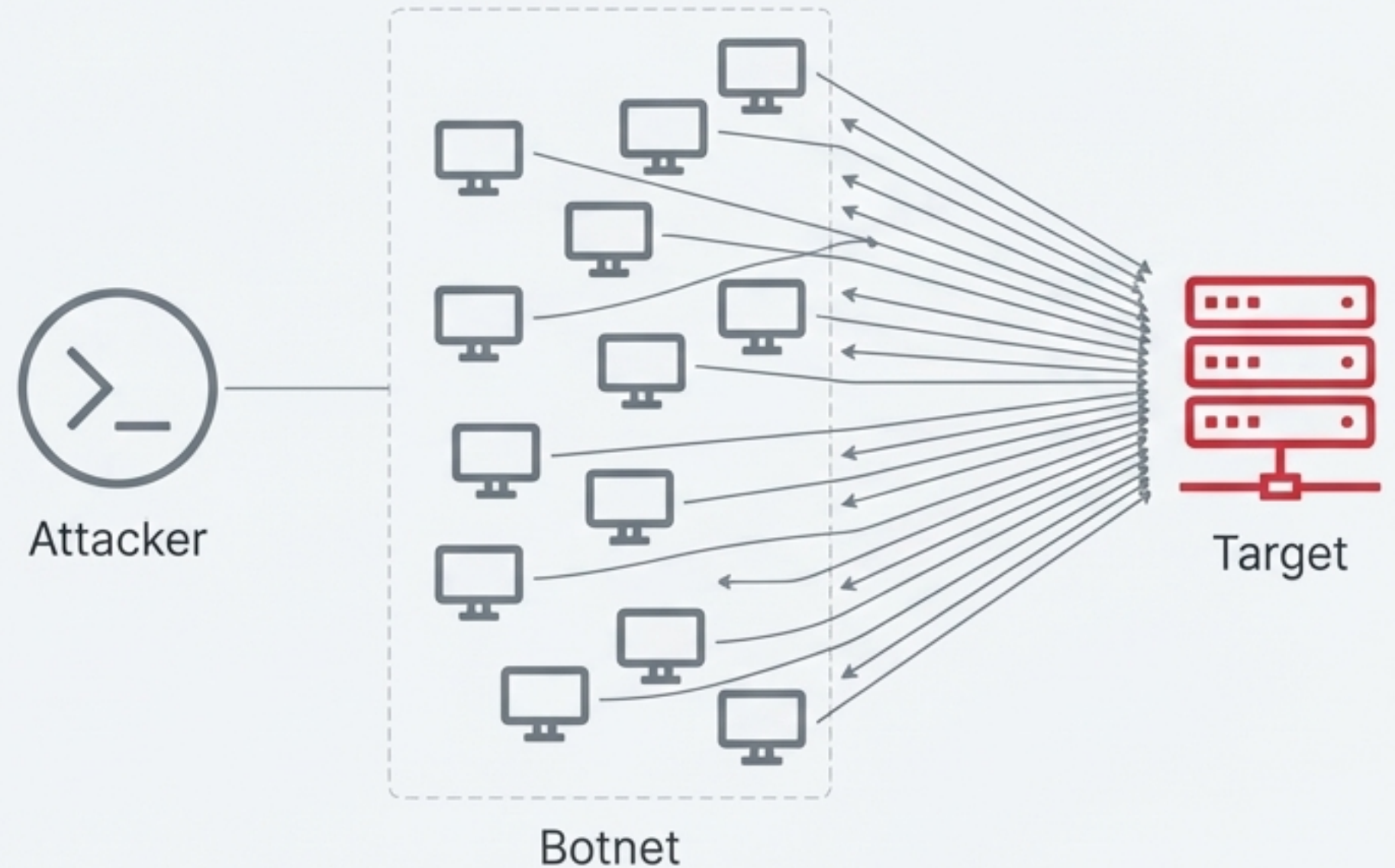


The Threat Scales: The Distributed Denial-of-Service (DDoS) Attack

A DDoS attack elevates the threat by using multiple sources to launch the attack simultaneously.

Attackers typically use malware to infect a large number of computers, creating a “botnet”—a network of compromised machines under their control.

This botnet is then directed to send an overwhelming volume of requests to the final target.

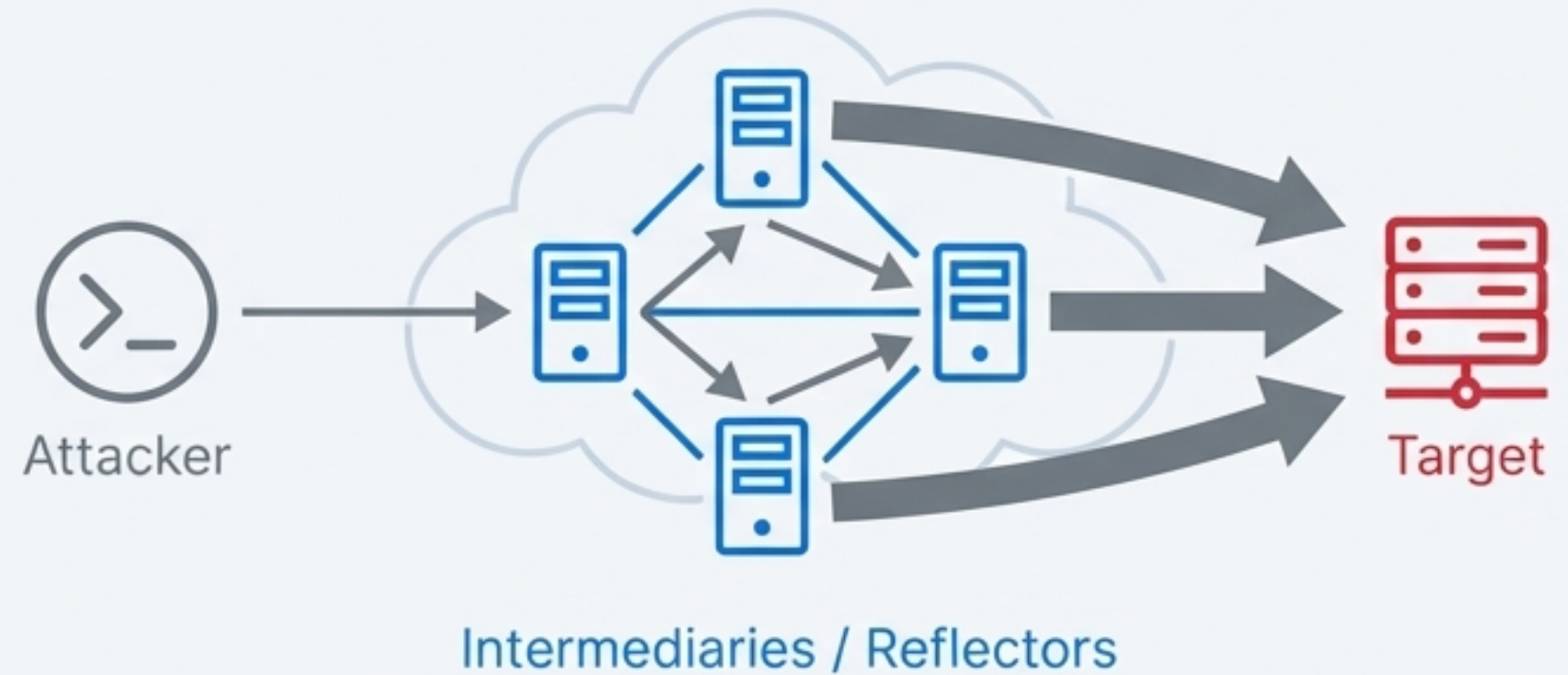


A More Deceptive Evolution: The Distributed Reflected Denial-of-Service (DrDoS) Attack

A DrDoS attack is a sophisticated type of DDoS attack that adds two crucial elements: **reflection** and **amplification**.

Instead of attacking the target directly, the attacker uses legitimate third-party servers on the internet as unwitting intermediaries.

This method not only hides the attacker's true origin but also magnifies the power of the original attack.



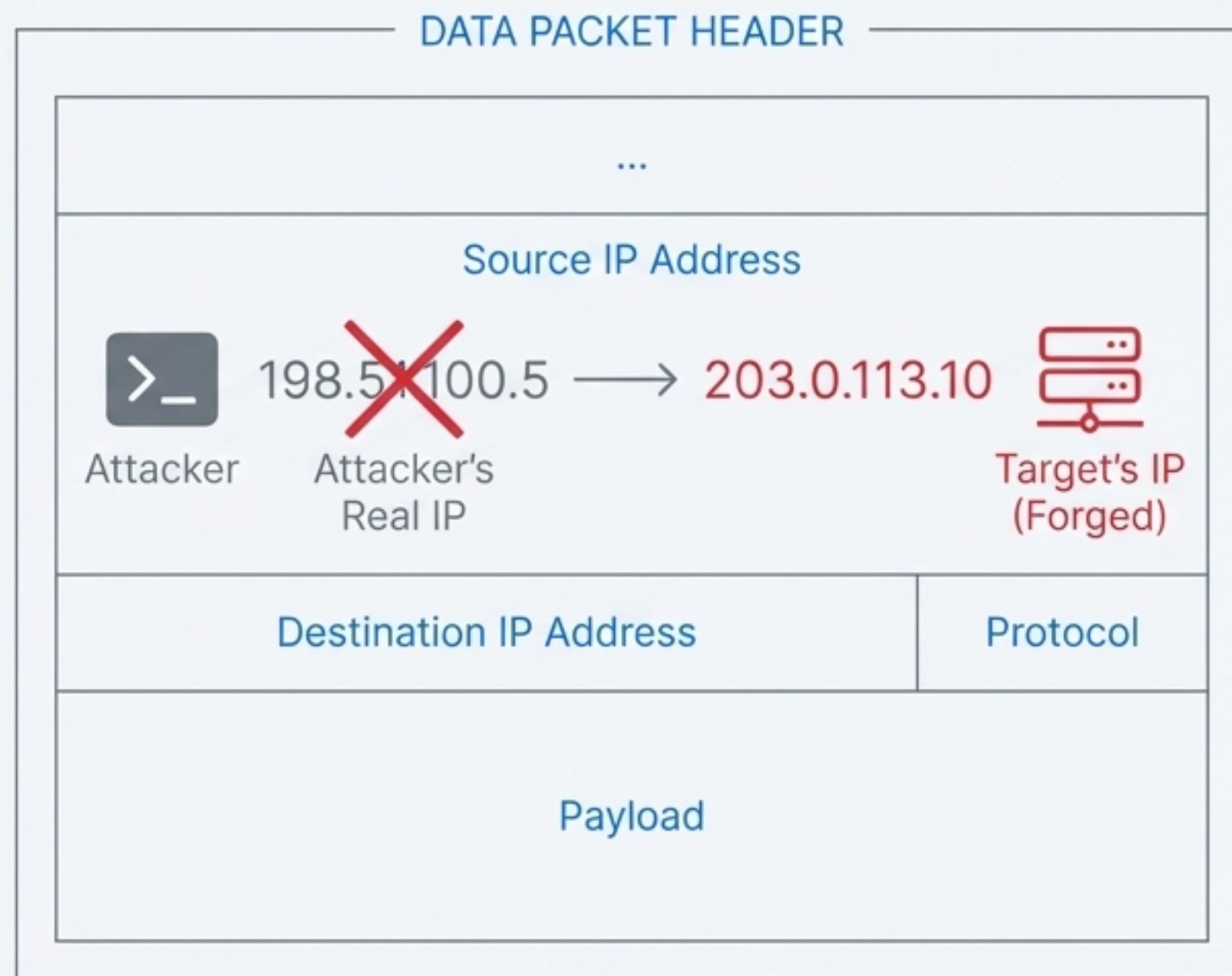
The Core Deception: Forging the Return Address with IP Spoofing

Every packet of data on the internet has a header, which includes a source IP address—the “return address”.

In a DrDoS attack, the attacker crafts request packets but forges the source IP address.

They replace their own IP address with the IP address of their **intended target**.

Any system that receives this packet will send its reply not to the attacker, but to the forged address of the target.



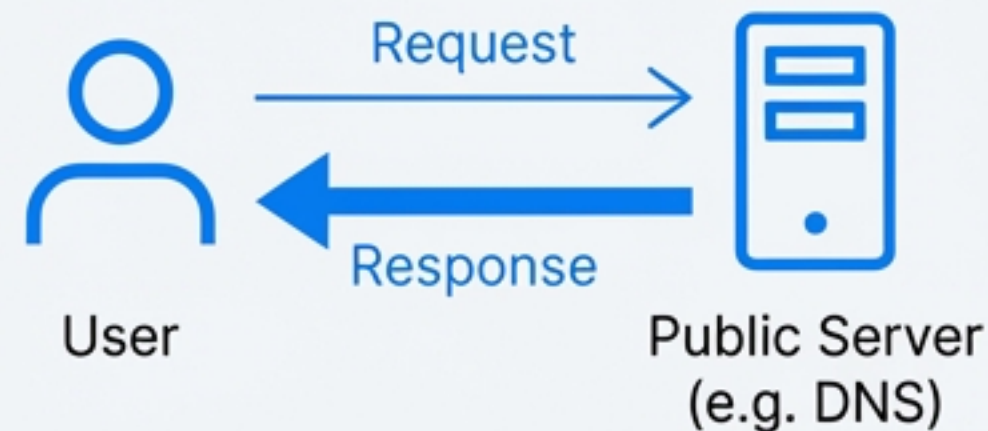
The Unwitting Accomplice: Exploiting Legitimate Servers for Reflection

The internet is built on request-response protocols. A client sends a request to a server, and the server sends a response back.

DrDoS attackers identify and select a large number of these publicly accessible servers (e.g., DNS, NTP servers) to act as “reflectors”.

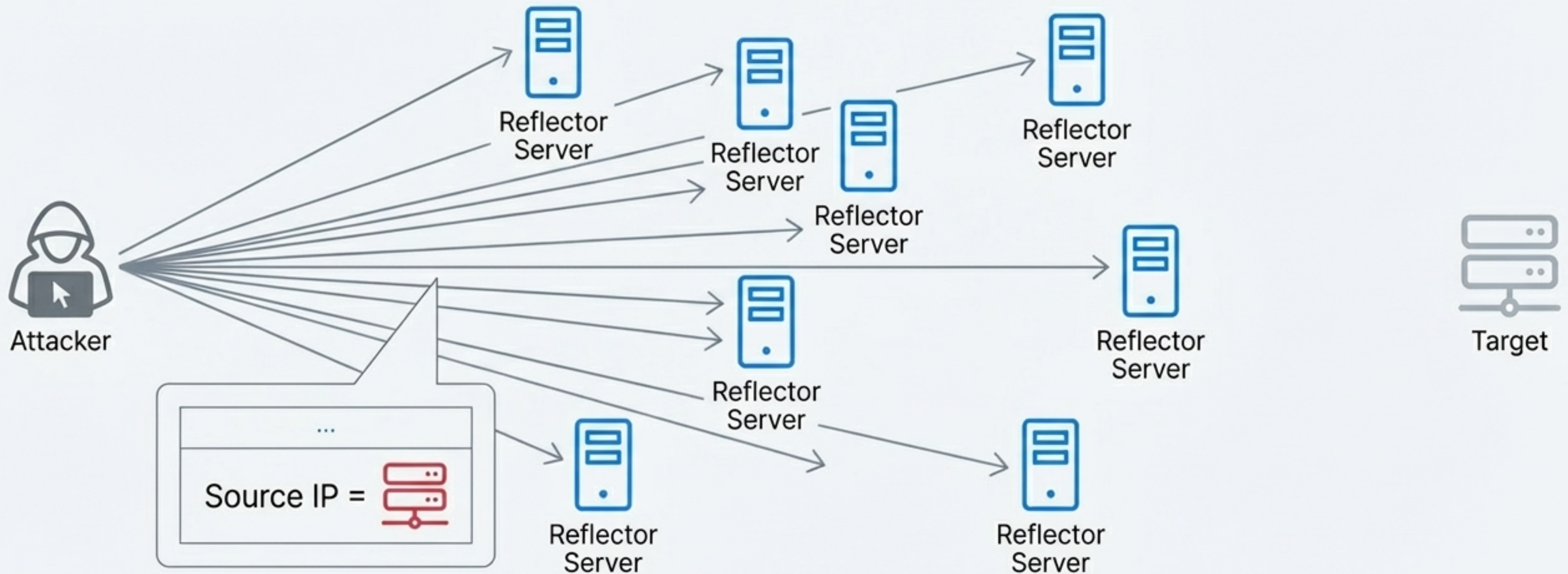
These servers are not compromised; they are simply performing their normal function. They have no way of knowing the initial request was sent with a forged source IP.

Normal Request-Response Cycle



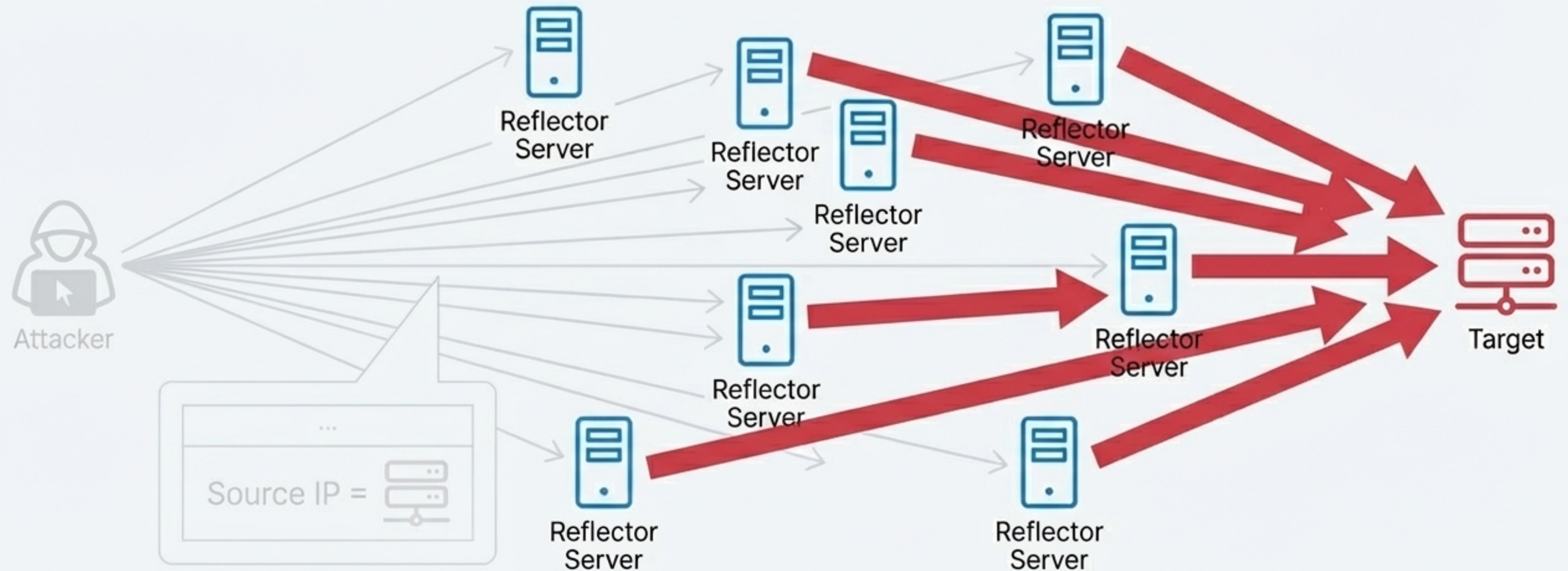
Anatomy of an Attack, Step 1: The Spoofed Request

The attacker sends a massive number of small request packets to numerous reflector servers. Critically, the source IP address on every one of these packets has been spoofed to be the target's IP address.



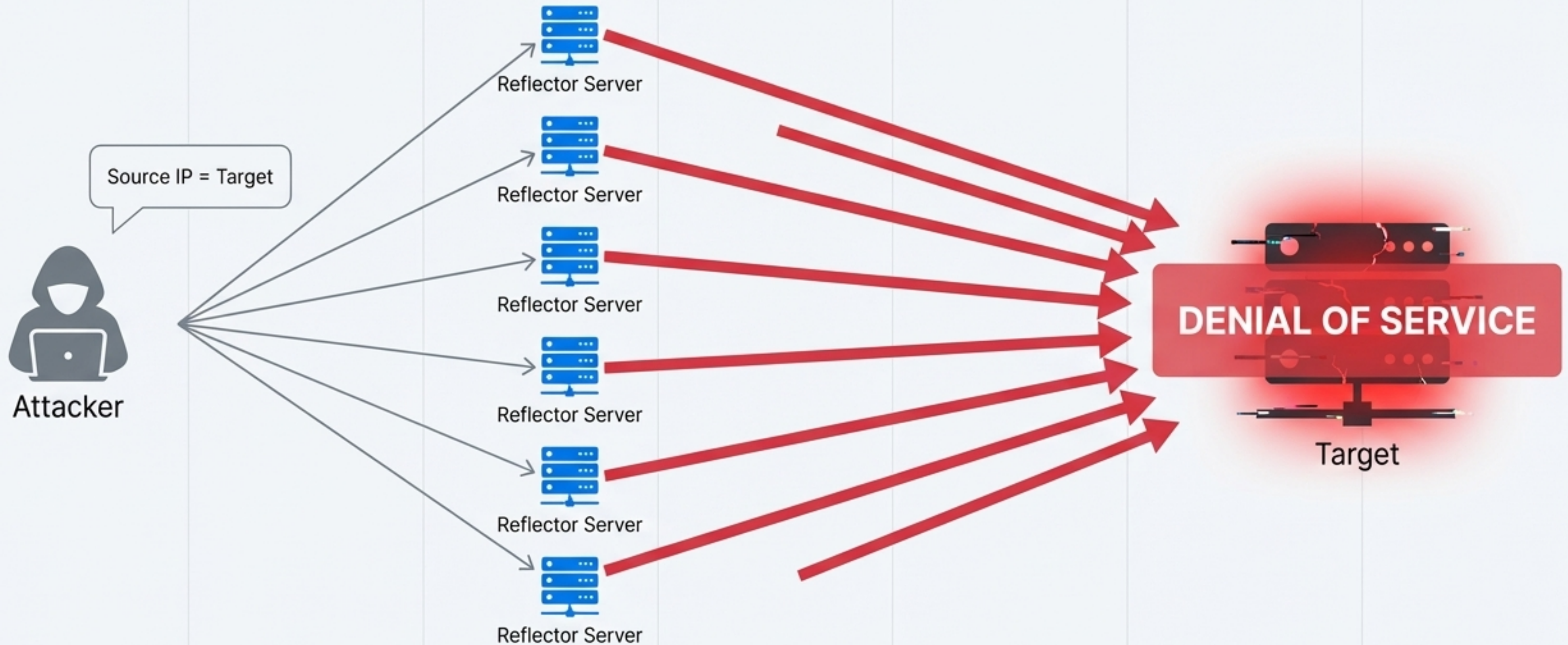
Anatomy of an Attack, Step 2: The Reflected Response

The reflector servers receive the requests. Behaving as designed, they generate responses and send them to the source IP address specified in the requests—which is the address of the unsuspecting target.



Anatomy of an Attack, Step 3: The Overwhelming Flood

The target is inundated with a massive volume of unsolicited response traffic from all the reflector servers. This flood of data consumes the target's network bandwidth and resources, rendering it inaccessible to legitimate users and causing a denial of service.

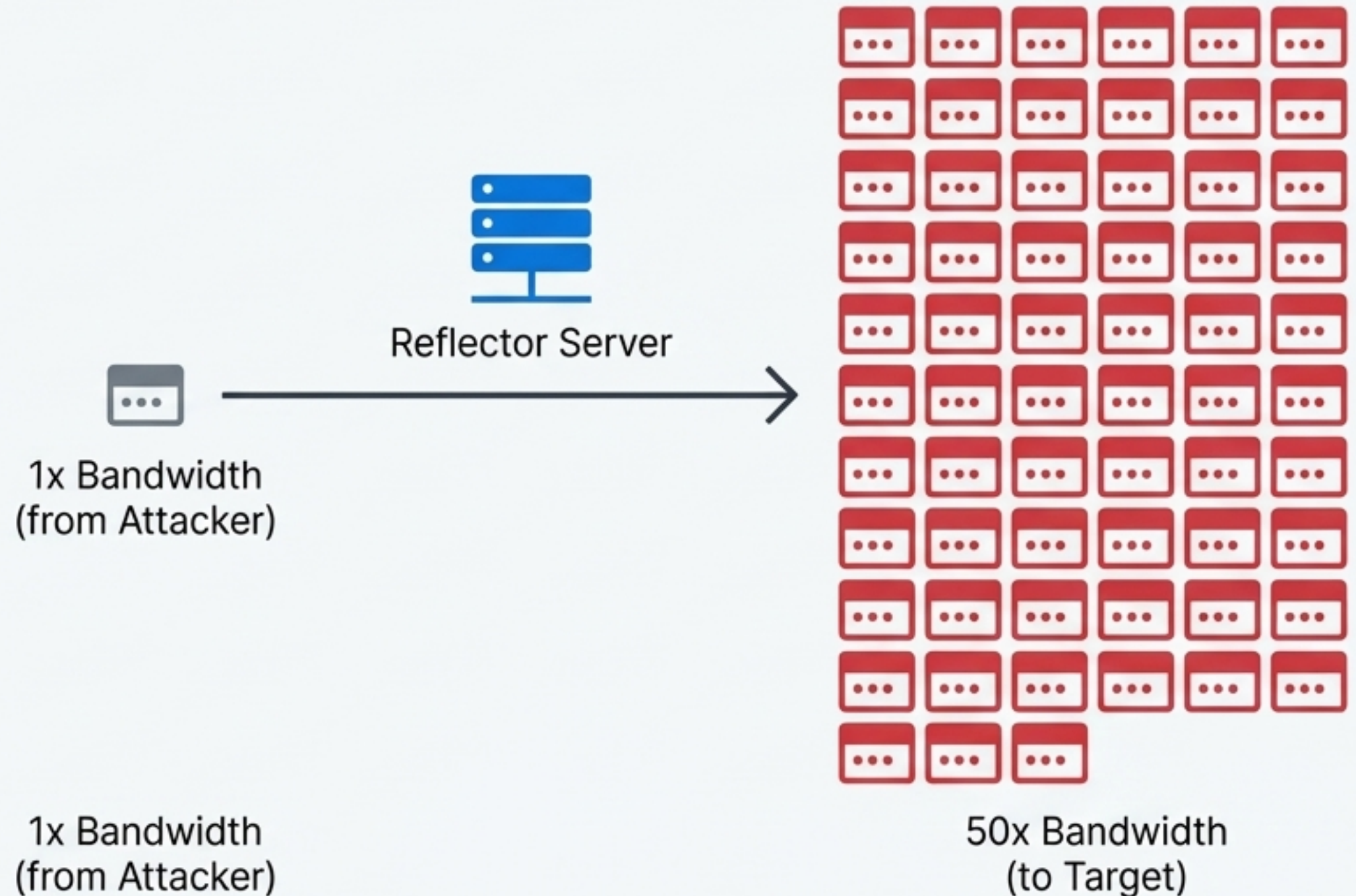


The Force Multiplier: The Amplification Effect

The primary advantage of DrDoS is amplification. The attacker can use a small amount of their own bandwidth to generate a much larger attack.

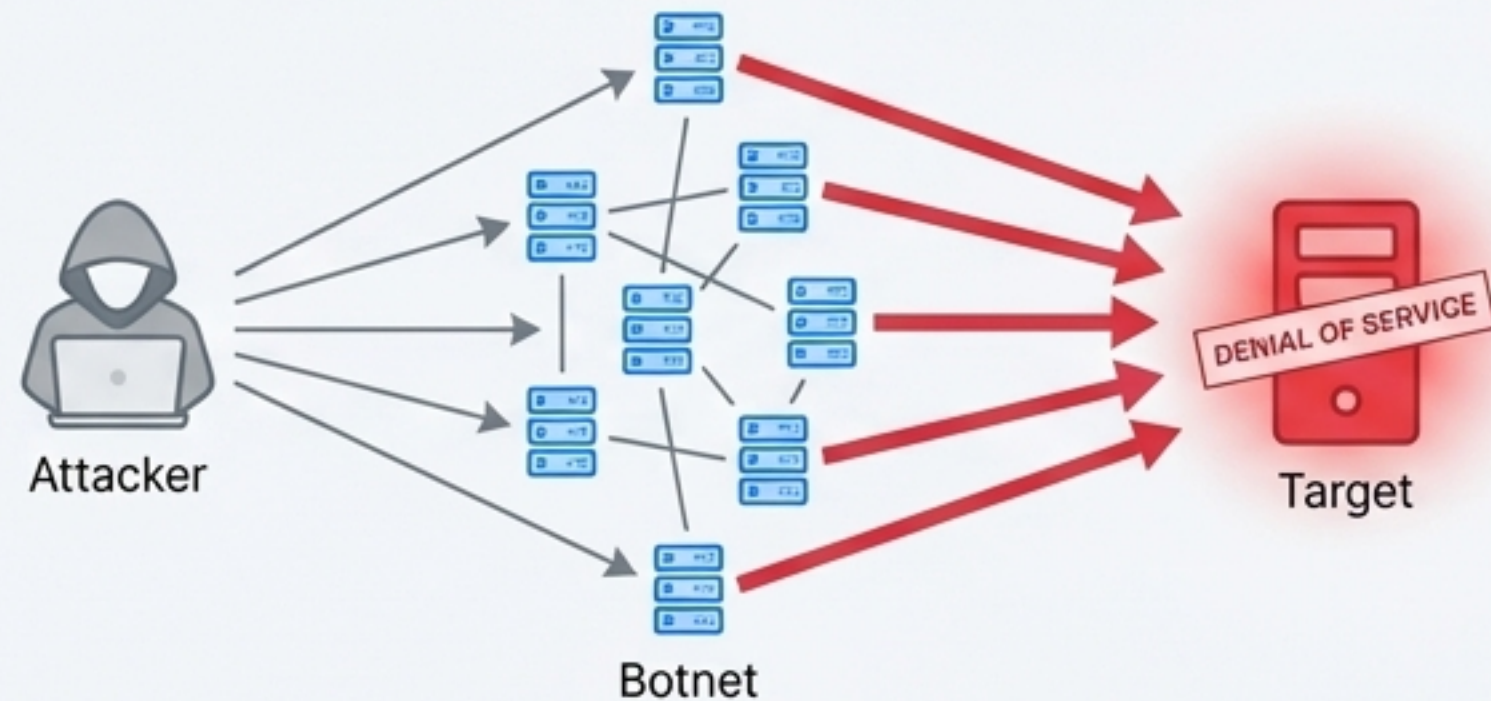
This is achieved by using protocols where the server's response is significantly larger than to initial request.

For example, a small DNS query can result in a response that is 50-70 times larger. The attacker sends the small query; the target receives the huge response.



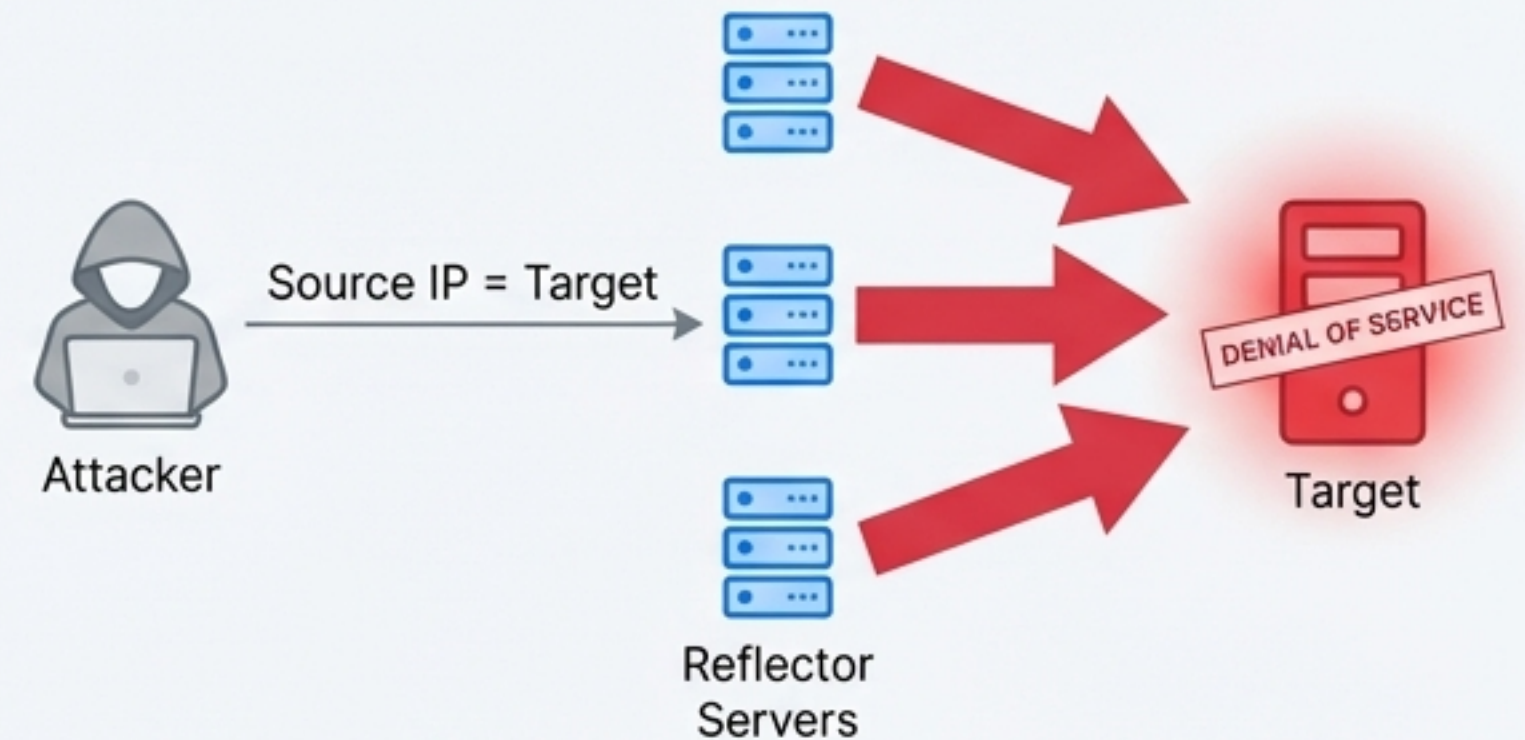
Key Differences: DDoS vs. DrDoS

Traditional DDoS



- Attacks the target directly from a botnet.
- Attacker's botnet is the source of the traffic.
- Attack size is limited by the total bandwidth of the botnet.

Reflected DrDoS



- Attacks the target indirectly via reflectors.
- Source of traffic appears to be legitimate servers.
- Attack size is amplified, far exceeding the attacker's own bandwidth.

Core Concepts of a DrDoS Attack



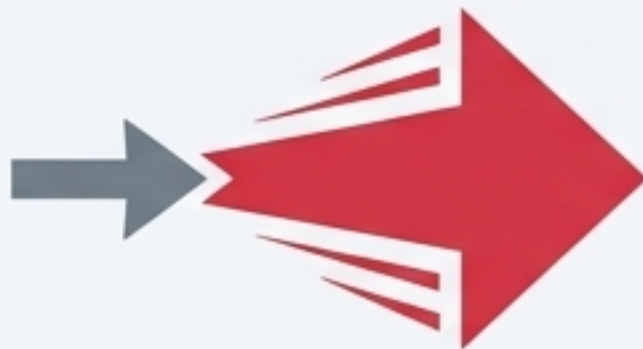
IP Spoofing

The attacker forges the source IP to misdirect responses to the target.



Reflection

Legitimate servers are used as unwitting intermediaries to bounce traffic.



Amplification

The response traffic sent to the target is many times larger than the attacker's initial request traffic.



Distribution

The attack traffic originates from a large number of reflectors, making it difficult to block.